Roll No. 22001025602

61478

B. Pharma 6th Semester (Regular/Re-appear/Improvement) Examination – May-2025

BIOPHARMACEUTICS AND PHARMACOKINETICS

Paper : BP-604T

Time : Three hours] *[Maximum Marks : 75*

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is ***compulsory***, each carry **2** marks, Attempt any ***two*** questions out of three from Section-**B**, each carry **10** marks and ***seven*** questions out of ***nine*** from Section-**C**, each carry **5** marks.

SECTION – A

1. Attempt all questions : $10 \times 2 = 20$

(a) How absolute bioavailability can be calculated ?

(b) Name any ***two*** protein involved in drug binding.

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(c) Differentiate between carrier mediated transport and passive diffusion.

(d) How urine pH Plays an important role in renal excretion of drugs ?

(e) What is enterohepatic circulation ?

(f) What is compartment model ?

(g) Write the significance of bioequivalence studies.

(h) What is the effect of increased Vd on loading dose ?

(i) Define accumulation ratio.

(j) What is Zero order elimination kinetics ?

SECTION – B

Note : Attempt any ***two*** questions : $2 \times 10 = 20$

2. Explain kinetics of protein binding in detail. Write a note on tissue permeability of drugs.

3. Differentiate between First order and zero order elimination kinetics. Enumerate factors causing non-linearity.

4. How does loading dose for a particular drug calculated ? Discuss kinetics of multiple dosing in detail.

SECTION – C

Note : Attempt any ***seven*** questions : $7 \times 5 = 35$

5. Explain in detail about plasma level time curve for a drug that follows a two compartment model system.

6. Illustrate trapezoidal method for the calculation of AUC.

7. How bioavailability can be measured using urinary excretion study ?

8. Write a note on kinetics of protein binding.

9. Define the following terms-elimination rate constant, Half-life and absorption rate constant.

10. Discuss the significance of steady state drug levels in clinical setting.

11. Describe briefly about nonlinear pharmacokinetics.

12. Write a note on non-compartment model.

13. Discuss various factors affecting renal excretion of drugs.

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2:00

Department of Pharmaceutical Sciences, I.G.U., Meerpur, Rewari
B. Pharm 6th sem, Biopharmaceutics & Pharmacokinetics
Ist sessional Exam, March 2025
(BP 604T)

Max Marks: 30 **Time: 1 hr**

Attempt all questions from part A (5x2)
Attempt 2 questions from part B (2x5)
Attempt 1 question from part C (1x10)

Part-A

(a). Area Under Curve (b) Volume of Distribution (c) Active transport (d) Elimination rate constant (e) Non compartment models

Part B

(a). Discuss various factors affecting protein drug binding.
(b). Explain Kinetics of Protein drug binding.
(c). Briefly describe various factors affecting drug absorption

Part-C

(a). Define the term absorption. Describe in detail mechanisms of drug absorption through GIT.
(b). Give determination of various pharmacokinetic parameters (KE, t1/2, Vd) in one compartment IV Bolus administration.

Department of Pharmaceutical Sciences

B. Pharm. 6th Semester, 2nd Sessional Examination, April, 2025

Paper Code – BP604T; Biopharmaceutics and Pharmacokinetics (Theory)

Total marks: 30

Duration: 1 hr 30 minutes

1. Give a brief description of the following: **(5×2=10 marks)**

a. ~~Loading and maintenance dose with formula~~ Membrane Excretion.

c. Name the agents used to estimate GFR for renal function.

d. Define 4 methods for enhancement of dissolution rate of poorly soluble drugs.

e. USP type III & IV dissolution apparatus

f. Pulmonary excretion of drugs

2. Write a note on the following (any two): **(2×5=10 marks)**

a. Define metabolism. Explain Phase II reactions.

b. Describe the factors affecting renal excretion of drugs.

c. Explain *in-vitro in-vivo* correlations with different levels.

3. Write a note on the following (any one): **(1×10=10 marks)**

a. Define bioavailability. Write its objectives. Explain different methods for measurement of bioavavilability.

b. Explain non-linear pharmacokinetics, factors causing non-linearity and Michaelis-menton method of estimating parameters.